

Bio-Inspired Algorithms: An Introduction (minicourse)

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1 Objectives

This one-week minicourse aims to introduce the main concepts and algorithms related to bio-inspired computing.

2 Program

Overview of Bio-Inspired Algorithms. Evolutionary Computing (The generic evolutionary loop: representation, fitness, selection, variation, replacement). Genetic Algorithms (Crossover, mutation, elitism; schema theorem; Evolution Strategies; Evolutionary Programming; Differential Evolution). Collective Intelligence (Swarm Intelligence; Ant Colony Optimization; Particle Swarm Optimization). Cellular Automata (Elementary automata; Wolfram's four classes; the Game of Life; evolving CA rules). Artificial Immune Systems (Self/non-self discrimination, tolerance, clonal selection, affinity maturation, idiotypic networks. Shape space and affinity; Negative selection; Anomaly and intrusion detection). Neural Computation (Hebbian learning; the perceptron; Hopfield associative memories; spiking models; neuroevolution). Python Programming.

Bibliography

- [1] Bonabeau, E., Dorigo, M., Theraulaz, G. (1999). *Swarm Intelligence: From Natural to Artificial Systems*. Oxford University Press.
- [2] de Castro, L. N. (2006). *Fundamentals of Natural Computing*. Chapman & Hall/CRC.
- [3] de Castro, L. N., Timmis, J. (2002). *Artificial Immune Systems: A New Computational Intelligence Approach*. Springer.
- [4] Dorigo, M., Stützle, T. (2004). *Ant Colony Optimization*. MIT Press.
- [5] Eiben, A. E., Smith, J. E. (2003). *Introduction to Evolutionary Computing*. Springer.
- [6] Goldberg, D. E. (1989). *Genetic Algorithms in Search, Optimization, and Machine Learning*. Addison-Wesley.
- [7] Haykin, S. (1999). *Neural Networks: A Comprehensive Foundation*, 2nd ed. Prentice Hall.
- [8] Holland, J. H. (1975). *Adaptation in Natural and Artificial Systems*. University of Michigan Press.
- [9] Mitchell, M. (1998). *An Introduction to Genetic Algorithms*. MIT Press.
- [10] Wolfram, S. (2002). *A New Kind of Science*. Wolfram Media.

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